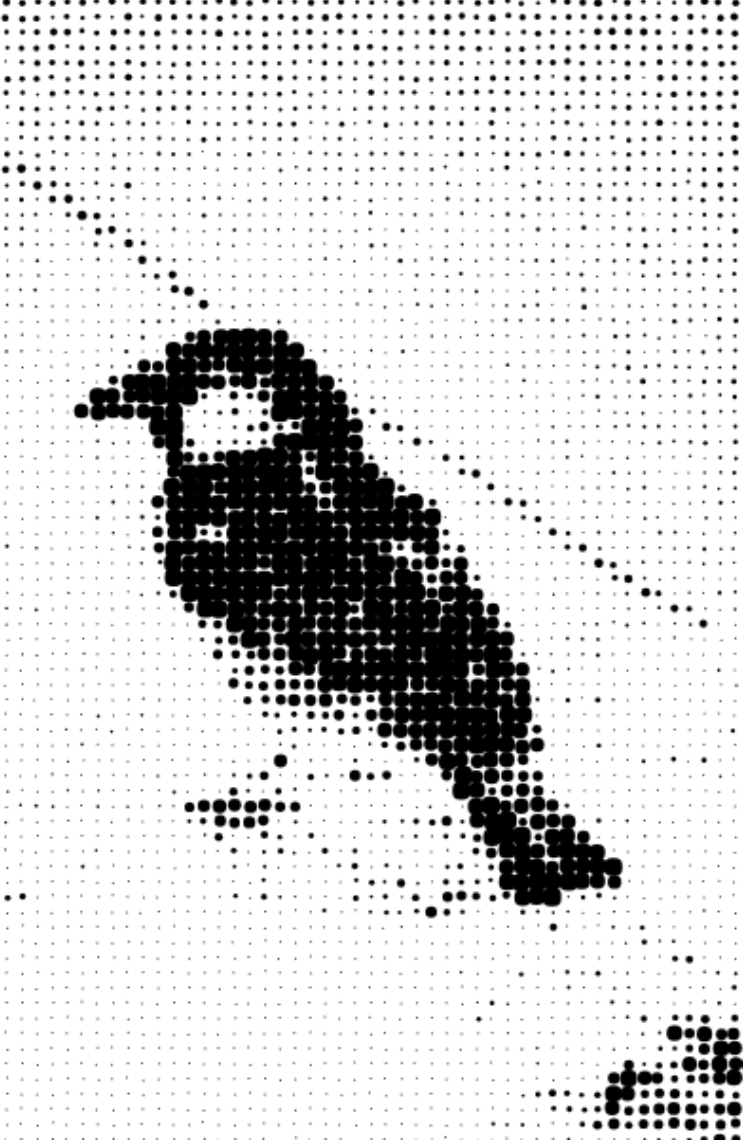

Rendering the Invisible

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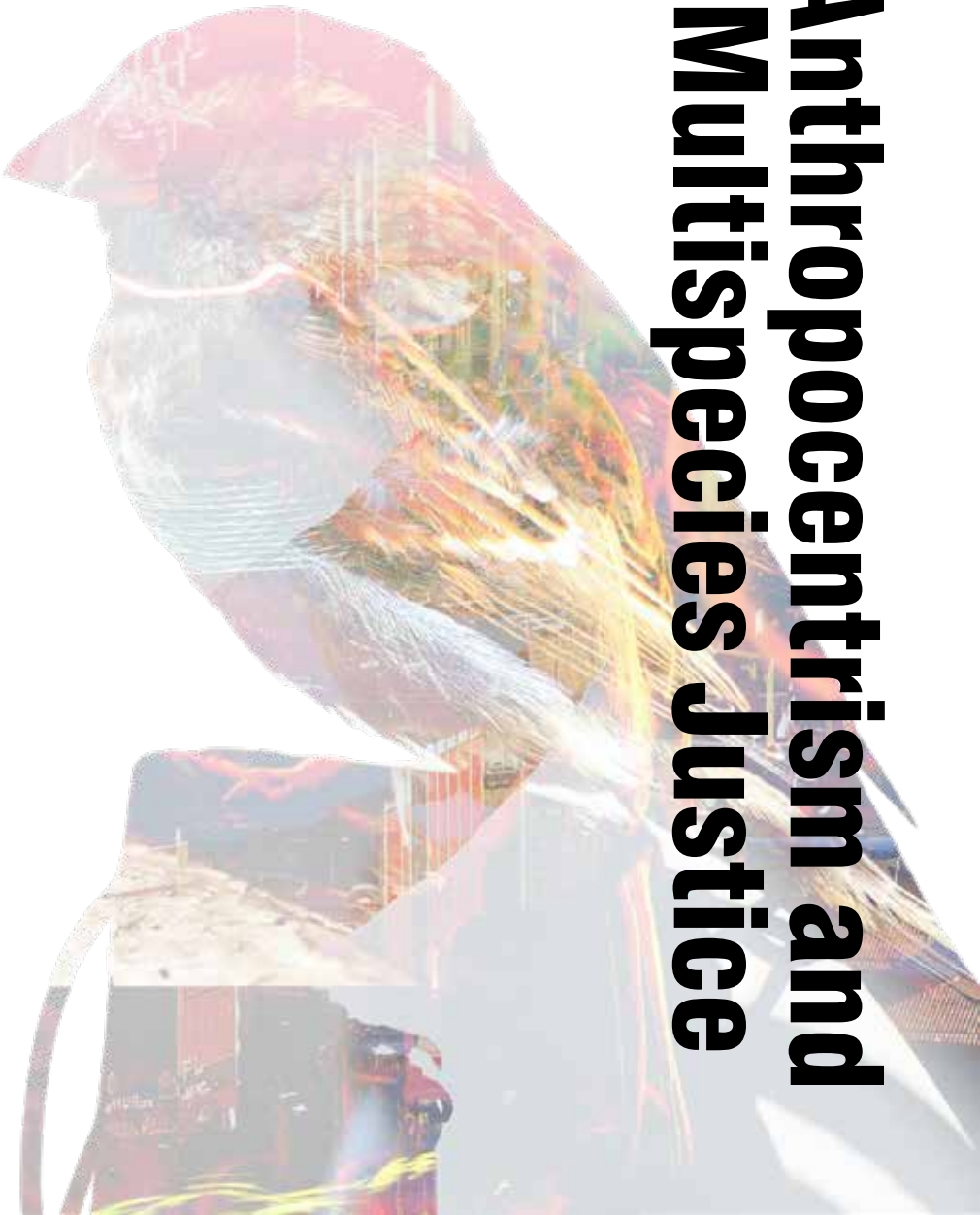
Anthropocentrism is a worldview that prioritizes human needs, comfort, and productivity in urban design, valuing other life forms only when they serve human interests (1,2). This perspective is not abstract but is embedded in urban infrastructures, planning policies, architecture, and economic systems, influencing what is considered a “problem” and what remains invisible (1).

Historically, modern Western thought has separated “nature” from “culture,” placing humans in a position of perceived dominance over ecological processes (3). This dualism has shaped cities into environments largely optimized for human efficiency; for instance, streets are designed for vehicle speed, buildings for maximum occupancy, lighting for human visibility, and soundscapes are calibrated to human tolerance (4). Non-human species are often accommodated only if they do not interfere with these anthropocentric objectives (1).

01

Anthropocentrism and Multispecies Justice



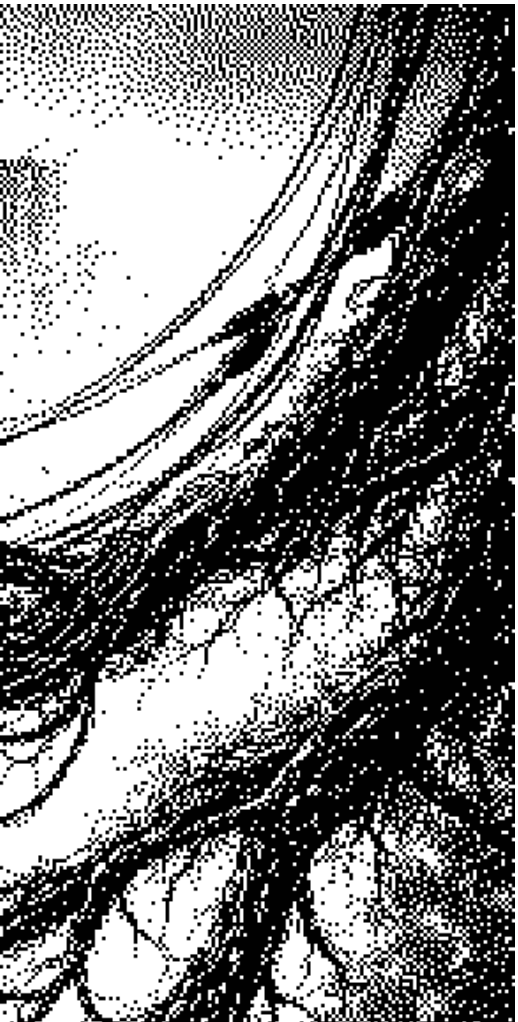
Urban research increasingly documents the ecological consequences of this human-centered calibration. Urban expansion leads to habitat fragmentation and loss of natural green spaces (4,5). Artificial light at night (ALAN) disrupts the circadian rhythms of various organisms, including birds and insects (6,7). Traffic noise interferes with animal communication, such as mating calls and alarm signals (8). Air pollution, including particulate matter and toxic emissions, can negatively affect the respiratory systems and overall health of urban wildlife (4,9,10). Because urban systems are often designed around human sensory thresholds, conditions that seem “normal” to humans can be biologically disruptive for other organisms (11).

This human-centric normalization is reinforced culturally and psychologically. A “human/nature dualism” legitimizes separation and domination (2), when humans perceive themselves as external to ecological networks, environmental damage can become an abstract concept. For example, traffic noise may become background, artificial light a signal of safety, and heat from asphalt a seasonal occurrence. Pollution might be reduced to numerical data in a report. In this scenario, humans adapt, while other species must either adapt more rapidly, relocate, or face extirpation (4, 5, 12).

Multispecies justice extends this critique by arguing that if cities are shared territories, then urban systems distribute vulnerability not only across social groups but also across species lines (1,2). While environmental justice research has demonstrated that pollution, heat stress, and lack of green space disproportionately affect marginalized human communities, multispecies justice expands this framework by asking what justice means when urban decisions shape the conditions of life and death for animals, plants, and ecosystems (1,2).

Urban design intrinsically reorganizes food systems, light cycles, water flows, and acoustic environments (4). Artificial lighting disrupts nocturnal species (6). Noise pollution interferes with communication and reproduction (4,8). Habitat fragmentation isolates populations (4). These are not incidental side effects but structural outcomes of design priorities centered on human perception.

As Haraway (2016) suggests, a key approach is to “make kin,” recognizing humans as deeply entangled within multispecies networks (13). Tsing (2015) similarly demonstrates how survival unfolds through interspecies collaboration, even in damaged landscapes (3). Together, these perspectives reposition cities as multispecies assemblages rather than exclusively human infrastructures. This shift does not reject human needs but reframes urban space as political territory shared by multiple forms of life (1,2). When design consistently reduces the livability of urban habitats for certain species, that reduction becomes a matter of justice (1).



02

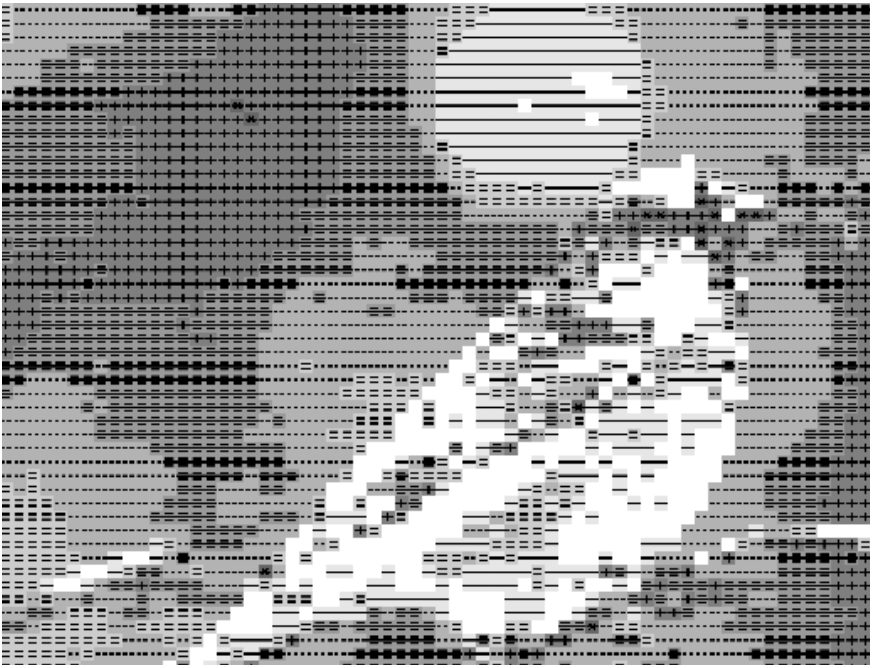
Umwelt and Sensory Invisibility

Jakob von Uexküll introduced the concept of Umwelt to describe the organism-specific perceptual world, which is shaped by an animal's sensory capacities and biological needs (14). For instance, a tick, a dog, and a human standing in the same physical space do not inhabit the same world; each experiences a different environment defined by what it can sense and respond to, highlighting that reality is relational (14). However, ecological research allows us to approximate aspects of another species' Umwelt by identifying measurable sensory thresholds and biological responses (11,15).

Sensory ecology builds on this idea by demonstrating how different species gather and interpret environmental information through various cues, including vision, sound, vibration, and chemical signals (15). Urbanization continuously reshapes these sensory environments. Streetlights alter nocturnal lightscapes (6). Traffic noise reshapes acoustic territories (8). Urban heat islands modify thermal perception (4,). Pollution alters chemical signals in the environment (16).

At the same time, humans experience a phenomenon of sensory normalization, where continuous stimuli become less perceptible over time. Robert Pyle's (1993) concept of the "extinction of experience" describes how reduced direct contact with non-human life diminishes our ability to perceive ecological change (17). Although Daniel Pauly's concept of shifting baseline syndrome is a relevant idea, it is not explicitly discussed in the provided sources.

Critically, what disappears from human awareness does not disappear in effect. Chronic urban noise can mask avian communication, leading to behavioral adjustments or reduced reproductive success (15). Artificial light at night disrupts circadian rhythms, affecting activity patterns and potentially hormonal regulation in urban birds (6). Accumulated heat and pollution act as physiological stressors for urban fauna, impacting their health and survival (4,5,9,18). The gap between human perception and non-human impact continues to widen.

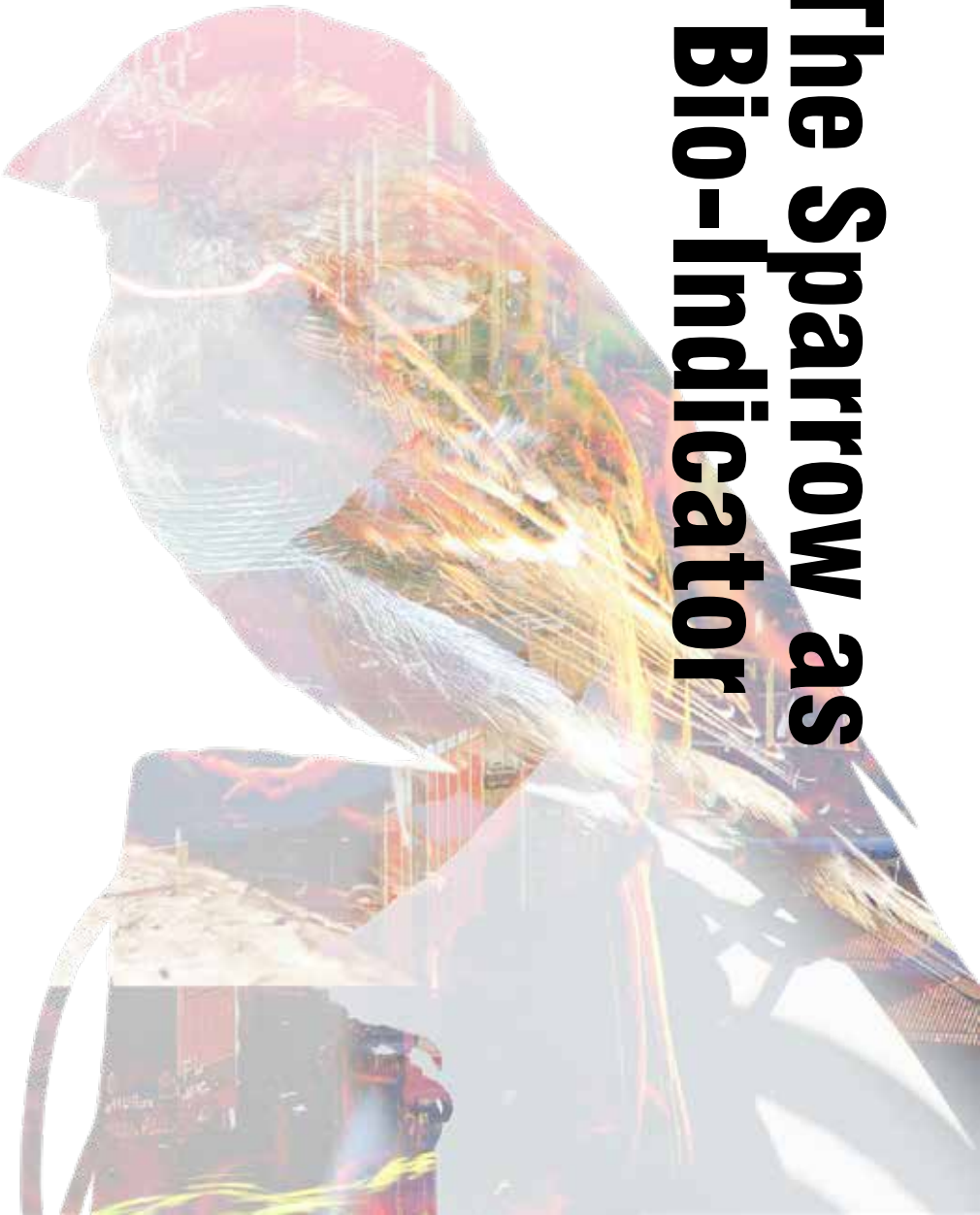


The house sparrow (Passer domesticus) offers a critical lens through which these urban dynamics become visible (9). Deeply integrated into European urban life, sparrows frequently nest in building cavities and forage in close proximity to humans (19). Their commonness often renders them background elements, yet this very proximity makes them ecologically revealing bioindicators of urban environmental health (16,20).

Long-term monitoring across European cities has documented significant declines in urban sparrow populations since the late twentieth century. Research in the Metropolitan Area of Barcelona confirms continued decreases in abundance, identifying factors such as loss of nesting sites, pollutant exposure, and food scarcity as principal drivers (12). This decline is cumulative rather than singular, resulting from multiple interacting stressors (10).

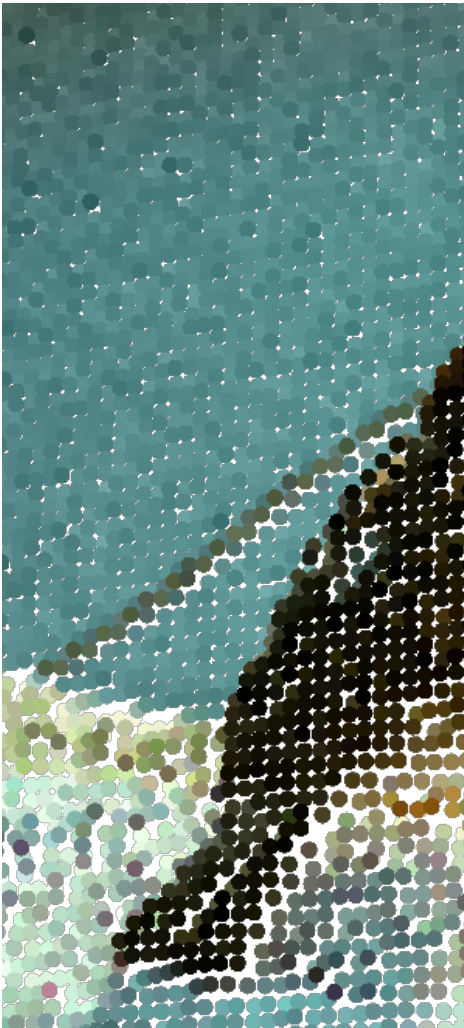
03

The Sparrow as Bio-Indicator



Food limitation: While adult sparrows can survive on seeds and human-provided food, nestlings require protein-rich insects for proper development (4,5,10,19). Reduced insect abundance in urban environments leads to lower fledgling survival and compromised reproduction (4,10,12).

Pollution exposure: Urban sparrows can exhibit elevated concentrations of heavy metals, such as copper, nickel, chromium, zinc, manganese, and cobalt, in their muscle, kidney, and liver tissues when compared to populations in unpolluted areas (16). This heavy-metal bioaccumulation is associated with altered morphometry, such as lower body mass and shorter wing/feather lengths, as well as oxidative stress and immune system activation (9).

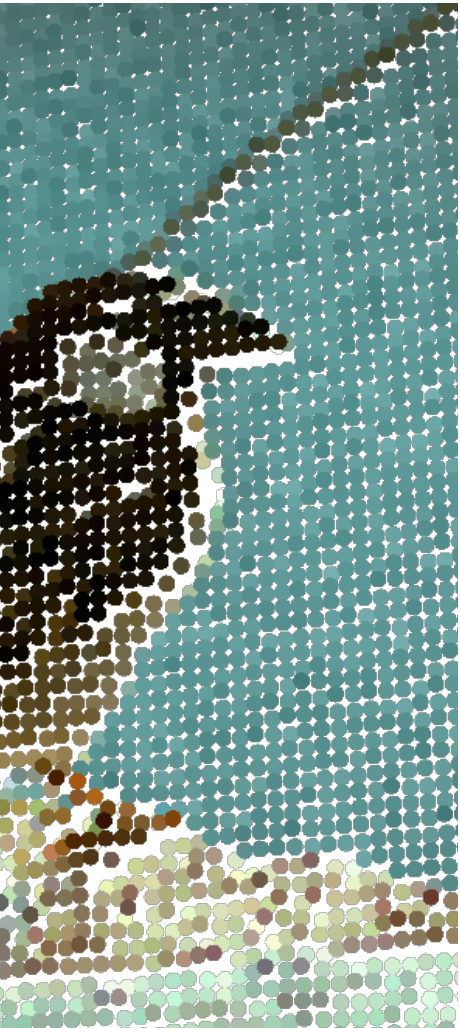


Nesting-site scarcity: As contemporary architecture prioritizes sealed façades and smooth surfaces, traditional crevices and roof gaps that sparrows historically used for nesting disappear (4,12). This loss of suitable nesting niches significantly reduces breeding opportunities for this obligate cavity-nesting species (4,12).

Noise and physiological stress: Low-frequency urban noise, primarily from vehicular traffic and industrial operations, overlaps with sparrow vocalizations, prompting frequency shifts or amplitude

increases (Lombard effect) to overcome masking effects (8). Elevated corticosterone levels and altered immune responses have been documented in sparrows from disturbed industrial environments, suggesting compromised physiological condition (5,18). Juvenile sparrows, in particular, show increased feather corticosterone concentrations, indicating developmental stress associated with urban conditions (11).

Artificial light at night: Exposure to artificial light at night (ALAN) can advance the daily onset of activity in house sparrows, particularly during pre-breeding and chick-rearing stages (6). While it may accelerate reproductive physiology in some avian species, chronic exposure to ALAN can disrupt hormonal regulation and sleep patterns, potentially leading to physiological costs (6,7).



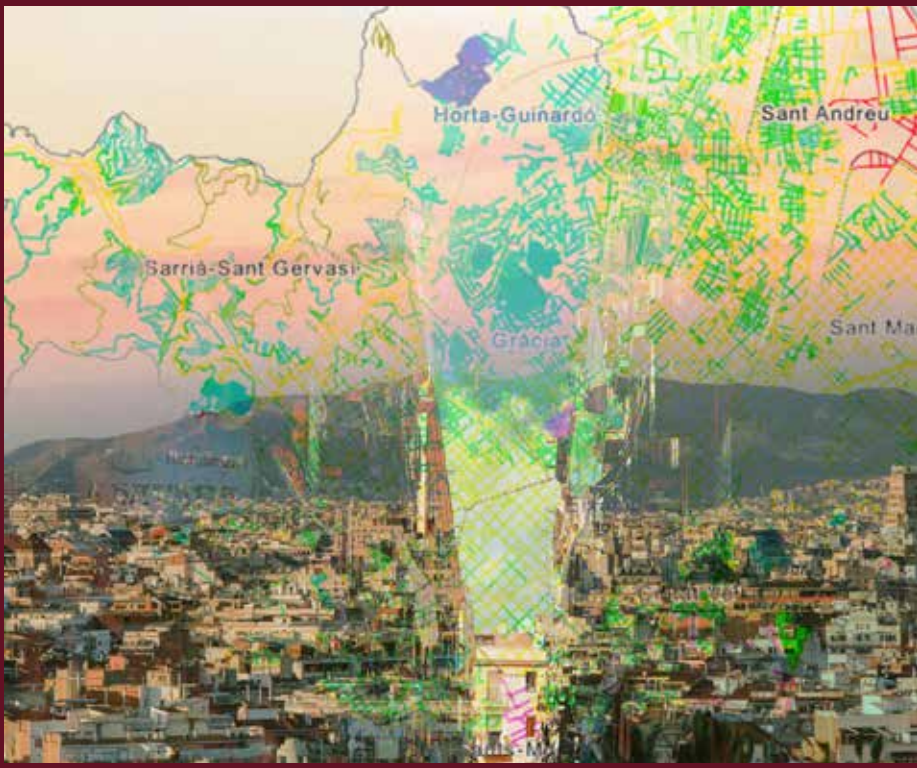
Although the house sparrow is globally classified as “Least Concern,” the species shows population reductions of up to 60% in several European urban centers (4,5,10,11,12). Its decline in dense urban cores therefore functions as a significant ecological signal of systemic environmental imbalance (4). Scientifically, the sparrow serves as a measurable bioindicator, and symbolically, it represents a species historically intertwined with human settlement whose vulnerability reveals the hidden costs of anthropocentric urban design (4,11,20).

04

Urban Context Barcelona

Barcelona provides a concentrated example of these dynamics, facing multiple environmental stressors that affect both human and non-human inhabitants (12,21). The metropolitan area, with a population recorded on January 1, 2025, by the Municipal Register of Inhabitants (22), is characterized by high building density, intensive traffic flows, and limited green space (5,12,21).

Air quality: Urbanization in European cities is associated with elevated levels of traffic-related air pollution and heavy metals (4). While specific data on PM2.5 and NO2 in Barcelona sparrows are not directly provided in the sources, studies show that urban sparrows in polluted contexts exhibit higher concentrations of heavy metals like copper, lead, and zinc in their tissues, reflecting chronic vehicular pollution exposure (16). This heavy-metal bioaccumulation contributes to oxidative stress and can reduce reproductive success (9,16).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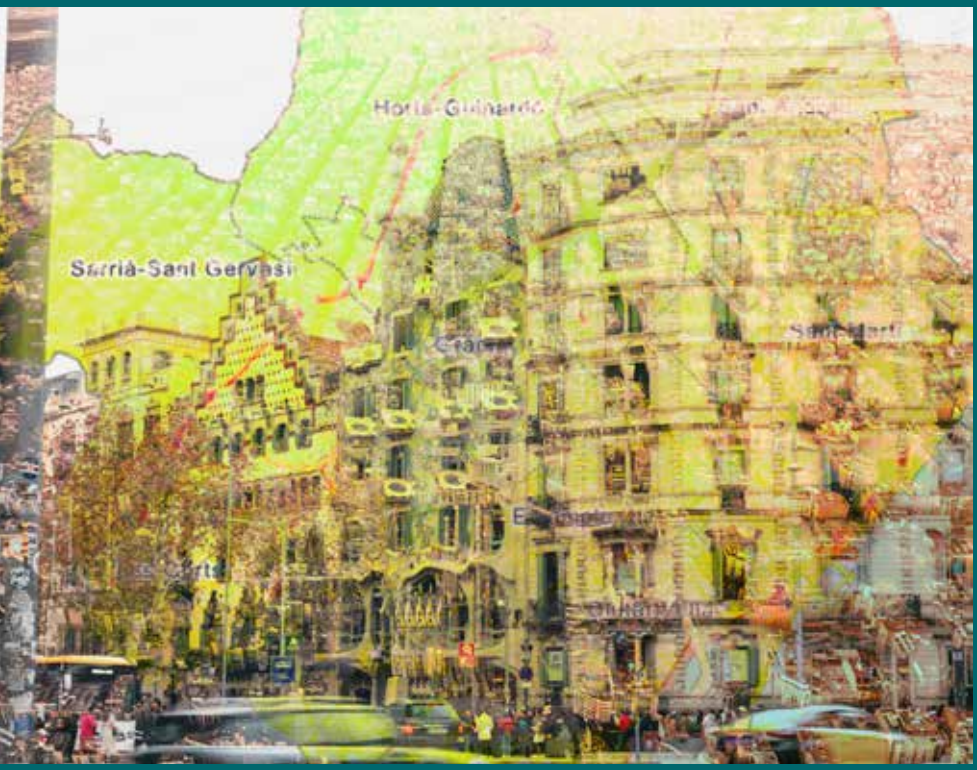
Noise: Road traffic constitutes a dominant environmental noise source in European cities (8). Municipal noise maps in Barcelona indicate persistently elevated dB(A) levels along major corridors (23). These noise levels can overlap with sparrow acoustic ranges, impairing communication and increasing physiological stress (4,8,15).

Urban Heat Island effect: Dense building blocks and impervious surfaces in urban areas elevate daytime temperatures and reduce nocturnal cooling, creating urban heat islands (4,11,24). For small passerines like sparrows, increased ambient temperatures can raise dehydration risk and metabolic demand (4).

Architectural typologies and habitat loss: Traditional low-rise buildings with façade crevices historically provided essential nesting opportunities for house sparrows (4,12,19). Contemporary sealed envelopes and modern architectural designs eliminate these micro-habitats (4,21). Habitat fragmentation and reduced vegetated patches, due to intensive urbanization, further limit sparrows' access to seeds and insect prey essential for chick development (4,12).

Mediterranean cities like Barcelona can sustain biodiversity, but rapid urban expansion fragments green areas and simplifies trophic networks (12). The decline in insect abundance, a critical food source, directly reduces food availability for sparrow nestlings, weakening long-term population stability (4,6,10,12).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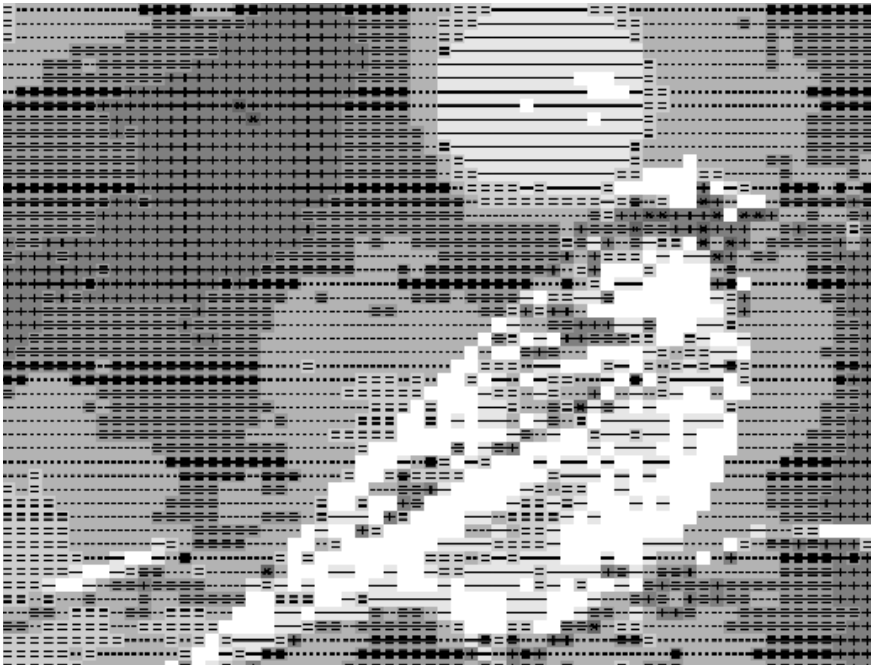
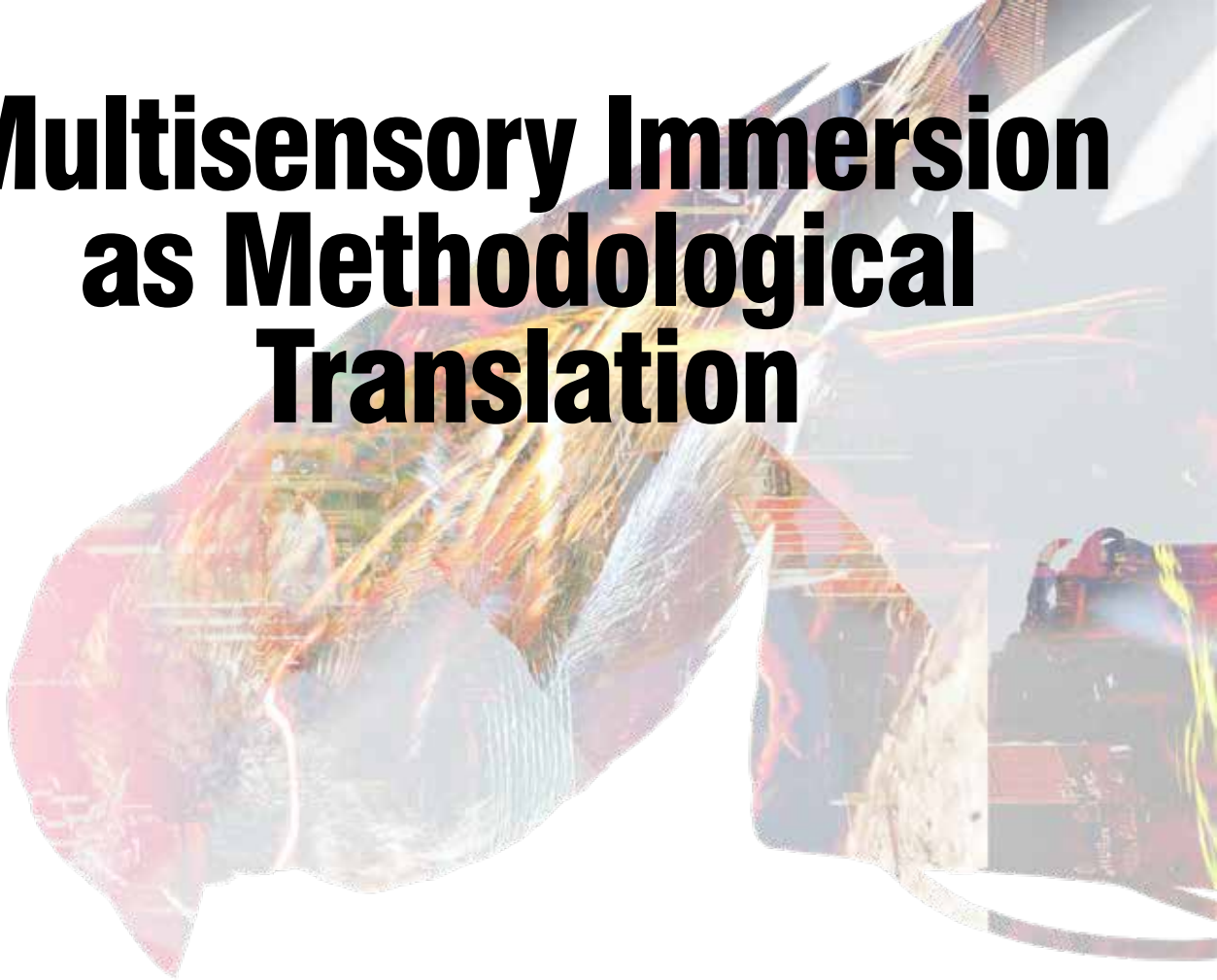


A multisensory immersive experience can function as a methodological bridge between ecological data and embodied human perception. The environmental stressors discussed throughout this research: anthropogenic noise, artificial light at night (ALAN), heat accumulation, and pollution, are typically communicated through abstract indicators such as decibel levels, lux measurements, temperature gradients, or particulate concentrations. While these metrics are scientifically rigorous, they often remain cognitively distant from lived experience. A multisensory installation offers a way to translate these variables into perceptible conditions without distorting their empirical basis (25).

Multisensory experiences are defined as immersive environments that simultaneously engage multiple sensory modalities, such as vision, audition, and touch, thereby increasing perceptual coherence and attentional focus (25).

05 Multisensory Immersion as Methodological Translation

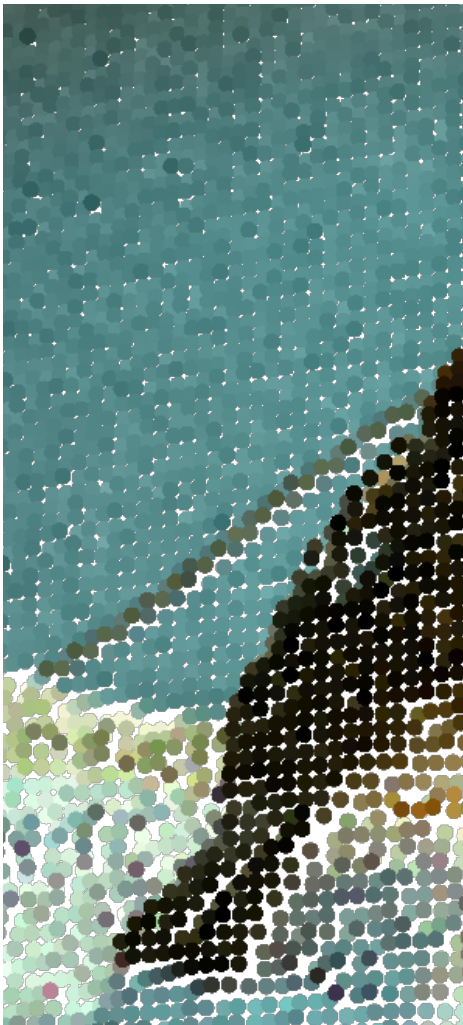


Research in cognitive science indicates that multisensory integration is a central feature of conscious perception; environmental stimuli are not processed in isolation but combined into unified experiential fields (26). Coordinating auditory, visual, and thermal cues therefore allows fragmented environmental stressors to be perceived as an interconnected system rather than as separate data points.

Immersion, in this context, is not a narrative embellishment but a structured perceptual condition. It aligns attention with a constructed environment designed to simulate specific sensory thresholds (27). Such environments can be evaluated using psychometric questionnaires, continuous subjective ratings, task-performance measures, and neuro-psycho-physiological indicators (27). These evaluation methods demonstrate that immersive systems can heighten presence, emotional salience, and attentional engagement when compared to single-modality representations (27).

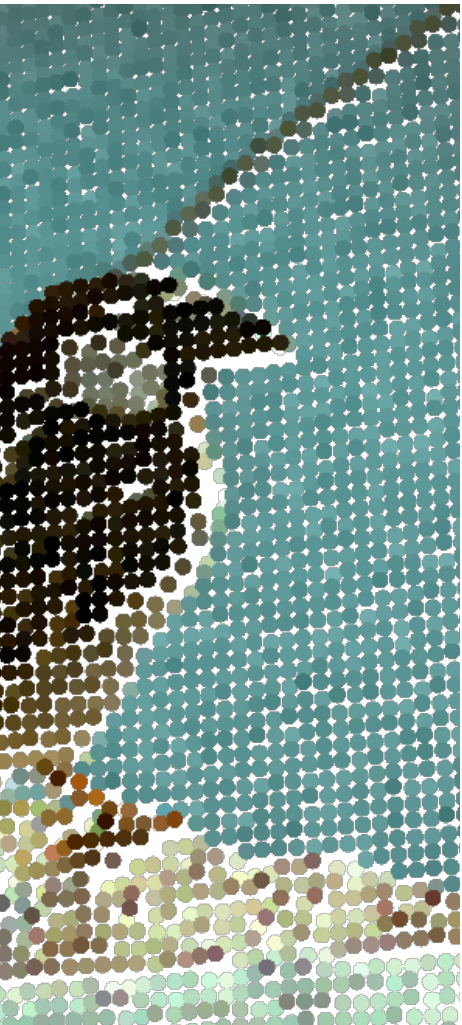
From a cognitive perspective, immersion narrows the distance between observer and phenomenon. Instead of interpreting environmental degradation through maps or statistics, participants encounter its sensory consequences directly. For example, the superimposition of low-frequency traffic noise onto birdsong can render acoustic masking perceptible (26). Dynamic lighting calibrated to mimic altered nocturnal spectra can approximate the visual impact of artificial illumination (26). When synchronized, these sensory cues create a perceptual field that reflects documented ecological pressures rather than symbolic representation.

Empirical research further suggests that multisensory immersion can enhance memory consolidation and emotional engagement, factors associated with shifts in environmental awareness and behavioral intention (27). Physiological measures such as heart-rate variability and skin conductance provide objective indicators of arousal during immersive exposure (27), supporting the claim that embodied engagement differs qualitatively from detached observation.



Within the framework of multispecies justice, immersive environments can re-scale anthropocentric thresholds. Decibel ranges, light intensities, and temperature differentials that fall within human tolerance can be recalibrated to approximate the sensory burdens experienced by other species (26). When grounded in documented ecological measurements, such recalibration does not constitute exaggeration but perceptual translation. This approach resonates with calls to “make kin” across species by fostering situated, embodied awareness rather than abstract empathy (26).

However, immersion should not replace ecological research.



Long-term monitoring, physiological analyses, and population studies remain the evidentiary foundation for understanding species decline (26). Immersive installations operate as communicative extensions of this research, spatializing and sensorializing established findings for broader audiences. When carefully calibrated to reflect real-world thresholds, multisensory immersion functions as perceptual amplification rather than aesthetic dramatization, strengthening both scientific communication and public engagement.

06

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